

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	ABCT4116
Subject Title	Synthetic Biology Project
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	NIL
Objectives	• Empower the students to build a better world by solving problems with the help of synthetic biology • Give students opportunities to push the boundaries of synthetic biology by tackling everyday issues facing the world • Bring students to work together to design, build, test, and measure a system of their own design using interchangeable biological parts and standard molecular biology techniques
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Conduct literature research and critically assess the reading materials; b) Demonstrate the ability in independent and creative thinking; c) Identify and solve scientific and technical problems; d) Formulate hypothesis, design and conduct studies, analyze and interpret data; e) Appreciate the importance of team work during experiment work, public promotion and organizing activities for the synthetic biology project; f) Manage time and organize efficiently; g) Communicate effectively during team work, public promotion, website writing and project presentation.
Subject Synopsis/ Indicative Syllabus	The project will involve different experimental investigations, also bioinformatics elements. Synthetic biology knowledge will be applied to initiate ideas, design experiments and evaluate bioprocesses. Conducting survey on the feasibility of the projects to field experts and promoting to the public about the ideas will be included as well.
Teaching/Learning Methodology	With guidance from the supervisor team, students as a group are encouraged to propose the project theme for iGEM competition. Students will be divided into different sub teams – dry lab / wet lab / production / integrated human practice in order to carry out different job duties to fulfil the prize criteria of the iGEM competition.

	Supervisors meet regularly with the students to monitor their progress to make sure the subject learning outcomes are being achieved. Students will also have regular meetings to keep track of the progress.								
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						
			a	b	c	d	e	f	g
	1. Class Participation / Class Work	30%	✓		✓	✓	✓	✓	✓
	2. Project proposals	10%	✓	✓	✓		✓		✓
	3. Assignments	10%	✓	✓		✓			
	4. Team work effort	30%	✓	✓	✓	✓	✓	✓	✓
	5. Project outcomes	20%	✓	✓	✓	✓	✓	✓	✓
	Total	100 %							
	Class participation, classwork and assignments of tutorials and seminars will be counted towards final grade. Students should be able to initiate an innovative synthetic biology project which has important impact on improving human’s life. They have to analyse and present the collected data results to the public. At the same time, students should promote synthetic biology by all means, but not limited to education, human practices and entrepreneurship approaches, to the public. Project outcomes are assessed based on their deliverables and the number of prize criteria the projects have met. The project is <u>assessed individually</u> by class participation, assignments, proposal writing, team work effort, and the completion rate of the project. The subject is offered as a Summer Term + Semester 1 (the following year) pattern, i.e. Summer Term extended to next Semester 1.								

Student Study Effort Expected	Class contact:	
	• Tutorial / meeting / seminar	15 Hrs.
	• Experiment classes	24 Hrs.
	Other student study effort:	
	• Preparation work (e.g. literature review, experimental design, public awareness work, assignments etc.)	100 Hrs.
	• Total student study effort	139 Hrs.
Reading List and References	Related books and journal articles.	